Does a Cointegration Filter Help in Forex Pairs Trading?

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Abstract

We ask whether an Engle–Granger–style tradability filter improves risk-adjusted foreign-exchange pairs trading relative to an always-trade rule. Using daily Yahoo Finance spot prices for seven major USD crosses from 2007 to 2025, we form all 21 undirected pairs and trade equal-notional spreads on rolling 257/21-day train/test windows. The filter rule trades an out-of-sample block only when a training-window residual ADF filter passes; the benchmark uses the same z-score rule without that gate. After a stylized 2 bp round-trip cost, the unlevered filter portfolio earns about 0.5% annualized at the pre-committed entry band $z^\star = \pm 1$ because capital is rarely deployed—the invested fraction is about 8% of pair-days—yet it delivers higher Sharpe and Sortino ratios than always-trade. A paired circular block bootstrap finds a statistically significant Sharpe gap ($\Delta = +0.59$, $p = 0.014$) and Sortino gap ($\Delta = +0.68$, $p = 0.018$) at ± 1; raw-return gaps are insignificant, and wider bands keep the ranking in point estimates without statistical separation. The ± 1 risk-adjusted edge survives a cost grid from 0 to 5 bp. The letter transfers equity-style pairs-filter horse races to liquid G10 FX; it is not a new cointegration estimator, and residual ADF passes are not claimed as size-correct cointegration detections at 5%.

Keywords: Cointegration; Pairs trading; Forex; Mean reversion; Sharpe ratio
JEL: C58; F31; G11; G15

Introduction

Pairs trading exploits temporary divergences between related assets (Gatev et al., 2006; Krauss, 2017). Distance and correlation filters treat mean reversion as a short-horizon regularity; cointegration reframes the problem by requiring a stationary linear combination of non-stationary prices, so that deviations from equilibrium are statistically constrained (Engle & Granger, 1987; Vidyamurthy, 2004). Equity applications of cointegration and related selection filters are well documented (Burgess, 2003; Galenko et al., 2012; Huck & Afawubo, 2015; Rad et al., 2015). Parallel evidence for major currencies is thinner, even though G10 USD crosses share a common numeraire and are linked by trade, capital flows, and parity relations that make long-run co-movement economically *plausible* (Dunis et al., 2011; Engel & West, 2005; Fama, 1984; Rogoff, 1996). Purchasing-power parity implies that related nominal rates are not free to wander independently forever, even when real exchange rates mean-revert only slowly (Rogoff, 1996). Covered and uncovered interest-rate parity connect spots, forwards, and interest differentials; even when uncovered interest parity fails as a short-horizon forecast (Fama, 1984), exchange rates remain asset prices tightly tied to macro fundamentals (Engel & West, 2005). Those links motivate *where* we look; they do not, by themselves, deliver a trading rule, and we do not estimate PPP baskets or trade UIP residuals.

This letter runs a clean selection-method horse race in liquid FX. We compare (i) a *simple* pairs strategy that always trades the equal-notional synthetic spread on every out-of-sample block with (ii) a *filter* strategy that trades only when an Engle–Granger–style tradability filter (Engle & Granger, 1987; Vidyamurthy, 2004) passes on the preceding training window. The design is deliberately close to equity filter evaluations such as (Huck & Afawubo, 2015): same entry logic with and without the gate, stylized costs, and portfolio-level risk-adjusted metrics. Ranking and forecasting approaches to pair selection (Huck, 2009; 2010) are complementary but outside our design. The contribution is transnational applicability of the filter logic to G10 FX—a setting that *Finance Research Letters* explicitly welcomes—not a new econometric procedure and not a single-country equity replication.

Our pre-committed headline design is $z^\star = \pm 1$ with a stylized round-trip cost $\kappa = 2$ bp of pair notional. At that design the filter book improves Sharpe and Sortino with formal portfolio-level inference, but the economic reading is a *filter* result: unlevered filter-book returns are small because occupancy is low. We do not claim large investable alpha. We also do not interpret residual ADF passes as MacKinnon-sized "cointegration detected at 5%" (MacKinnon, 1994; 2010); the residual cutoff is a pre-committed tradability tuning parameter in the sense of (Vidyamurthy, 2004). Open replication code and a frozen price panel accompany the letter.

Design

Data and windows

Daily closes for EUR, GBP, JPY, CHF, CAD, AUD, and NZD against USD come from Yahoo Finance (Yahoo Finance, 2025), 1 January 2007 to 31 December 2025 ($T = 4944$ calendar observations in the frozen panel). Quotes with USD as numerator are inverted so every series is XXXUSD. Analysis uses log prices. Missing values are forward-filled within the download pipeline and dates with any incomplete cross-section are dropped, so the seven series share a common calendar. Prices are processed in rolling blocks: a training window of $n = 257$ days (about one year) followed by a testing window of $m = 21$ days (about one month). After each test block, windows advance by m days so the next train ends where the previous test ended. Training samples estimate the hedge and the spread moments used in the out-of-sample z-score; the simple rule skips the cointegration gate but still uses those moments. The 257/21 split is pre-committed rather than optimized in-sample; we do not search over alternative window lengths in this letter.

Tradability filter

We gate trading with the Engle–Granger two-step procedure (Engle & Granger, 1987), used as a *tradability filter* rather than as a formal size-correct cointegration test. Johansen methods can recover multiple relations in larger systems; for bivariate FX pairs we keep the Engle–Granger blueprint for transparency.

Step 1. Ordinary ADF tests on each observed log-price level (constant; AIC lag). Both legs must fail to reject a unit root ($p > 0.05$).

Step 2. OLS long-run regression $y_t = \alpha + \beta x_t + \varepsilon_t$ with residual $\hat{\varepsilon}_t = y_t - \hat{\beta} x_t - \hat{\alpha}$.

Step 3. Ordinary ADF test on $\hat{\varepsilon}_t$. We pass the window when the residual ADF p-value is at most 0.05 and use $\hat{\beta}$ as the hedge ratio for the synthetic spread. Because β is estimated, that residual statistic does *not* follow the Dickey–Fuller distribution of Step 1: a size-correct Engle–Granger test would compare it against MacKinnon cointegration critical values, which are stricter (about -3.36 versus -2.87 at the conventional 5% level for $N = 2$ with a constant on our 257-day windows; (MacKinnon, 1994; 2010)). We deliberately keep the looser ordinary-ADF residual cutoff as a pre-

committed tuning parameter that controls how permissive the filter is. A pass means the residual looks stationary enough, under this operational rule, to trade the spread out of sample—not that cointegration was detected at 5%.

Cointegration is a relation, not a directed trading object, so we work with $C(7,2) = 21$ unordered pairs. Legs are ordered alphabetically by ticker. The residual filter can depend on which series is the regressand; as a fixed design choice we therefore run both OLS orientations on each training window and, among those that clear the residual cutoff, retain the orientation with the clearer residual ADF t-statistic (more negative), mapping that hedge into the alphabetical spread. If neither orientation passes, the pair is flat for the following test window.

Trading rule, costs, and portfolio

On each testing day the z-score of the spread is $Z_t = \frac{\hat{\varepsilon}_t - \mu}{\sigma}$, where μ and σ are the training mean and standard deviation of $\hat{\varepsilon}_t = \log P_t^1 - \beta \log P_t^2$. Enter a long spread when $Z_t < -z^\star$ and a short spread when $Z_t > z^\star$, for $z^\star \in \{1, 2, 3\}$. Signals are lagged two days. Pair PnL is equal-notional, $R_t = s_t(r_t^1 - r_t^2)$, not β-hedged. Flat days enter the Sharpe denominator as zeros.

Let $s_t \in \{-1, 0, 1\}$ denote the lagged signal. We charge a round-trip cost of κ basis points of pair notional on position changes, $\text{cost}_t = (\kappa/10^4) \times |s_t - s_{t-1}| \ /2$, so that opening or closing costs $\kappa/2$ bp and a long–short flip costs a full κ bp. Net returns are gross minus cost. We report $\kappa \in \{0, 1, 2, 5\}$, with headline results at $\kappa = 2$ bp—a stylized all-in haircut for liquid G10 spot, not a full bid–ask plus rollover model. Swap costs are left as a caveat.

The simple rule trades every testing window for every unordered pair. The filter rule trades a testing window only if the tradability filter passes on the preceding training window. Pair returns are averaged across the 21 pairs to form an unlevered 1/21 book. Occupancy is the mean *invested fraction*: the average across days of (number of pairs with a nonzero lagged signal)/21. Active days are the share of calendar dates with at least one live pair.

The 21 pairs are not independent experiments: every leg is a USD cross, and many pairs share a second currency. Pair-level Sharpes are therefore descriptive of the cross-section, not twenty-one separate hypothesis tests. Primary evidence is the portfolio comparison of filter versus always-trade. On those two daily series we test whether Sharpe and Sortino differences are distinguishable from zero with a paired circular block bootstrap that resamples calendar dates once and applies the same blocks to both books, preserving contemporaneous cross-book correlation (Ledoit & Wolf, 2008). Block length equals the 21-day test window ($B = 10\{,\}000$). We studentize the Sharpe difference with a Newey–West HAC standard error (Newey & West, 1987) (Bartlett; lag from the $4(T/100)^{2/9}$ rule) and report a percentile interval for Sortino under the paper's downside definition (standard deviation of strictly negative daily returns). A HAC delta-method Sharpe check (Jobson & Korkie, 1981; Memmel, 2003) agrees with the bootstrap. Calmar ratios remain descriptive: maximum drawdown is path-dependent. The threshold ± 1 is the pre-committed headline design; ± 2 and ± 3 are reported alongside it.

As an implementable companion, on each training block we estimate pair spread volatility and size every live pair on the following test block to 3% annualized train-window volatility, using the same leverage factor on both books; the companion book is the sum of pair PnLs, not 1/21. The 3% cap is an arbitrary choice. Cumulative-return figures separately rescale the filter path so that ex-post daily volatilities match the simple path; those figures are visual aids only.

Results

Table 1 reports the headline unlevered book at $z^\star = \pm 1$ after $\kappa = 2$ bp. The filter strategy earns 0.49% annualized at 0.89% volatility (Sharpe 0.55; Sortino 0.63) versus −0.17% at 4.02% volatility for always-trade (Sharpe −0.04; Sortino −0.05). Occupancy explains the gap in scale: invested fraction is 8.2% for the filter versus 51.3% for always-trade, so most of the 1/21 book is cash on a typical day. Active days are 68% versus 95%. Maximum drawdown is −2.5% versus −15.9%. Frictionless Sharpes at ±1 are 0.62 (filter) and 0.01 (simple).

The risk-adjusted improvement is statistically separable at this design: ΔSharpe = +0.59 ($p = 0.014$; 95% studentized CI $[+0.12, +1.07]$) and ΔSortino = +0.68 ($p = 0.018$; 95% percentile CI $[+0.12, +1.28]$). The annualized mean-return gap is not significant ($p = 0.33$). That split is the filter story: the filter's edge shows up in the volatility denominator, not in a large return differential.

At ±2, filter versus simple Sharpes are 0.41 versus 0.06, with invested fractions 3.0% versus 19.2% and maximum drawdowns −1.9% versus −12.5%. At ±3, Sharpes are 0.48 versus 0.10, invested fractions 0.95% versus 5.7%, and drawdowns −0.86% versus −9.3%. The Engle–Granger gate itself is on for about 18.5% of pair-days at every threshold; always-trade fills almost all out-of-sample blocks. Bootstrap ΔSharpe p-values are 0.145 and 0.089 at ±2 and ±3, and ΔSortino p-values are 0.270 and 0.332, so wider bands keep the ranking in point estimates without conventional statistical separation. We therefore treat ±1 as the inferential headline and wider bands as descriptive robustness, consistent with the pre-committed design rather than a post-hoc threshold search.

Across $\kappa \in \{0, 1, 2, 5\}$, ΔSharpe at ±1 stays between +0.57 and +0.61 with $p \leq 0.015$; ΔSortino stays between +0.65 and +0.69 with $p \leq 0.021$. At ±1, frictionless Sharpes are 0.62 (filter) and 0.01 (simple); at 5 bp they are 0.45 and −0.12. Block lengths of 5, 21, and 63 days leave the ±1 conclusion unchanged, as does dropping the initial warm-up zeros before the first completed train/test cycle.

Metric	**Filter**	**Always-trade**
Ann. return (%)	0.49	−0.17
Ann. volatility (%)	0.89	4.02
Sharpe	0.55	−0.04
Sortino	0.63	−0.05
Calmar	0.20	−0.01
Max. drawdown (%)	−2.48	−15.89
Invested fraction (%)	8.24	51.27
Active days (%)	68.33	94.58
ΔSharpe (p)		+0.59 (0.014)
ΔSortino (p)		+0.68 (0.018)

Table 1: Unlevered 1/21 portfolio at $z^\star = \pm 1$ after $\kappa = 2$ bp round-trip costs (257/21 windows; 21 undirected pairs; 2007–2025). Δ rows are filter minus always-trade from a paired circular block bootstrap ($B = 10\{,\}000$; block length 21). Occupancy is independent of κ.

Figure 1 shows equal ex-post volatility cumulative returns at ±1 after $\kappa = 2$ bp. The filter path rises more steadily than the always-trade path, which suffers deeper early-sample drawdowns. The figure is a visual aid only: the main claim is the unlevered portfolio comparison in Table 1.

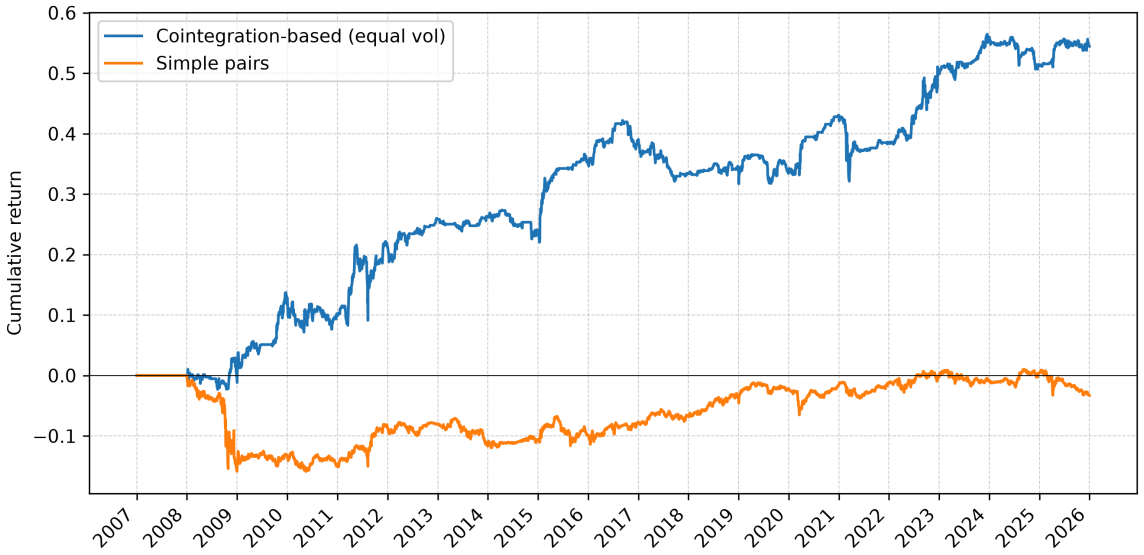


Figure 1: Cumulative returns at $z^\star = \pm 1$ after $\kappa = 2$ bp, with the filter path scaled to equal ex-post daily volatility versus always-trade (visual comparison only).

Sizing each live pair to 3% train-window volatility raises the filter companion to about 3.9% annualized return at about 8.3% book volatility (Sharpe 0.47) versus 1.9% return at 22.0% volatility for always-trade (Sharpe 0.09). Book volatility follows how many pairs are live under a common per-pair rule; the companion does not force equal occupancy.

Two caveats keep the claim honest. First, Yahoo spots are a convenience sample; we treat the exercise as a ranking horse race under a frozen public panel, not as executable prime-broker PnL. Absolute return levels could move under a professional vendor, but the object of interest is whether the filter ranks above always-trade after the same costs and the same portfolio construction. Second, equal-notional PnL does not match the estimated hedge ratio used to form the spread; β-hedged returns are left to companion work. Overnight swaps are outside the stylized κ grid; short test windows and frequent flats limit overnight exposure relative to an always-on book, but they do not eliminate financing frictions. Flats mechanically compress volatility, which is why invested fraction and active days belong beside Sharpe in the main table: without occupancy, a reader can mistake a low-vol filter for a high-capacity strategy. Replication materials (frozen parquet, `uv` pipeline, and bootstrap script) are linked in the data-availability statement so that the ΔSharpe and ΔSortino tests can be recomputed from the same panels.

Conclusion

In liquid G10 FX, gating an equal-notional pairs rule with an Engle–Granger–style tradability filter improves Sharpe and Sortino at the pre-committed band $z^\star = \pm 1$ after stylized costs, with paired block-bootstrap support. Raw returns remain small because the filter keeps occupancy low: about 8% of pair-days are live at ± 1, versus about half the book under always-trade. Wider entry bands preserve the point-estimate ranking without statistical separation. A 3% per-live-pair volatility companion still ranks the filter first on Sharpe, at higher book volatility that scales with how many pairs are on. The letter transfers an equity-style pairs-filter evaluation to major FX; it does not propose a new estimator, and the residual ADF cutoff is an operational tradability rule rather than a MacKinnon-sized cointegration claim. Tightening that cutoff to formal Engle–Granger critical values, aligning PnL with the estimated hedge, and adding parity-based or Johansen filters are natural extensions for future work.

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Data availability: Daily FX spots and replication scripts are available at https://github.com/QuantLandi/fx-cointegration (frozen Yahoo-derived panel `data/fx_prices.parquet`; MIT code/data; CC BY 4.0 manuscript assets under `paper/`).

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